HelixQA Autonomous Sessions — HelixVPN nano-detail spec (Volume 8 · §11.4.169 type 6)

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> **Reconciled (§11.4.35, 2026-06-26):** the MVP Definition-of-Done set has **NINE** acceptance criteria (AC1–AC9) per the canonical overview `10-testing-acceptance-and-qa.md` §7.2. The earlier "8 ACs" wording (§2, §6) and the §4 `status.json` counters that totalled 8 (`pass:5 + pending:3`) were a mis-count that left AC9 uncovered — a §11.4.118 coverage bluff. Corrected to NINE (the snapshot now totals 9: `pass:5 + pending:4`).

> Nano-detail expansion of §5.6 of the Volume-8 overview. HelixQA (`HelixDevelopment/helix_qa`) is the **autonomous QA orchestrator** (§11.4.27 / §11.4.107 / §11.4.158): it drives the full MVP-DoD acceptance set end-to-end with **no human**, consumes the `challenges` banks (challenges.md), boots infra via `containers` (integration.md), drives the UI via `panoptic` and analyses frames via `vision_engine`, generates tickets for findings into the workable-items DB, respawns crashed sub-runs (§11.4.147), and emits the **real-time sync channel** (§11.4.116 — append-only JSONL event stream + atomically-rewritten status snapshot) that the conductor tails live. It is the §11.4.165 **independent-verification agent for the runtime layer** — structurally separate from the code author. This document fixes the harness, the written test banks per app/service/platform, the captured wire-evidence-per-check contract, the sync channel, the determinism,

> the gate, the paired §1.1 mutation, and skeletons. Spec-only; unproven assumptions marked `UNVERIFIED`. Siblings: unit.md, integration.md, e2e.md, full-automation.md, challenges.md.

Table of contents

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1. Scope — what HelixQA orchestrates

HelixQA is not a single test type — it is the **autonomous QA session orchestrator** that drives every other type (CHAL, E2E, INT, SEC, REC) as an end-to-end run with no human, captures wire evidence per check, and reconciles the verdicts into the coverage ledger and the workable-items DB. Per §11.4.27 it must be used with **written test banks (suites) for every application, service, and platform** plus **fully-autonomous QA sessions** driving every registered bank with captured wire evidence per check.

HelixQA responsibility	Mechanism	Constitution
drive the DoD acceptance set end-to-end, no human	`--suite mvp-dod` over the `challenges` banks	§11.4.27 / §11.4.98
boot infra for the run	`containers/pkg/boot` (rootless)	§11.4.76 / §11.4.161
drive the UI flows	`panoptic` (web/desktop/mobile) → window-scoped MP4	§11.4.154 / §11.4.159
analyse frames for the REC verdict	`vision_engine` OCR + core-FSM cross-check	§11.4.163 / §11.4.165
emit a real-time sync channel	append-only JSONL events + atomic status snapshot	§11.4.116

HelixQA responsibility	Mechanism	Constitution
respawn crashed sub-runs	durable agent registry, crash≠done	§11.4.147
generate tickets for findings	write into the §11.4.93/.95 workable-items DB	§11.4.148
be the independent verifier	structurally separate from the code author	§11.4.165

The **written test banks** are organised per surface:

Bank	Surface	Suites it holds
helix_qa/banks/control_plane.yaml	the Go services	enrollment, RLS, reconcile, revoke, events, API authz (drives INT + CHAL)
helix_qa/banks/data_plane.yaml	helix-core + edge	reach, default-deny, escalation, kill-switch, liveness (drives E2E + SEC)
helix_qa/banks/clients_<platform>.yaml	per-platform app	connect flow, status UI, exit picker — one per platform (Access/Connector/Console × iOS/Android/Linux/Windows/macOS/Web; HarmonyOS/Aurora Phase 3)
helix_qa/banks/mvp_dod.yaml	the release gate	the 9 ACs (challenges.md §8)

> **UNVERIFIED:** the exact bank-file naming and the per-platform split granularity are the intended layout; the obligation (a written bank per app/service/platform, §11.4.27) is fixed, the file layout is a Phase-1 instantiation per §11.4.35.

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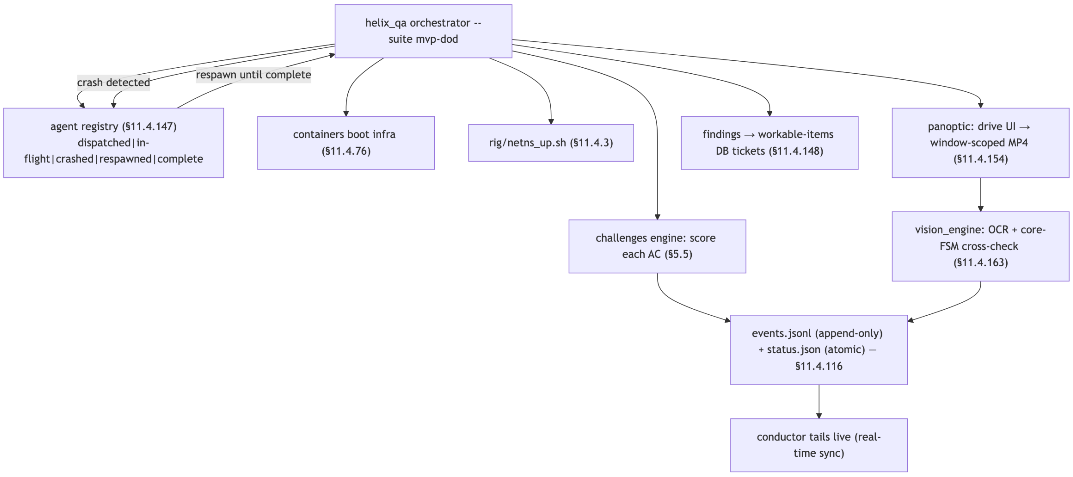
2. Harness — the helix_qa orchestrator + test banks

helix_qa (HelixDevelopment/helix_qa, Go) is consumed as an owned submodule (§11.4.28, equal-codebase, consumed by reference / pinned SHA), invoked as one command (overview §9): `go run ./submodules/helix_qa/cmd/orchestrator --suite mvp-dod`.



- **Independent verifier (§11.4.165):** HelixQA is dispatched as a structurally- separate agent from the code author — it re-reads the captured evidence and produces its own verdict; a HelixQA PASS is the runtime-layer independent confirmation, never the author asserting their own work.
- **Crash-respawn (§11.4.147):** every sub-run (per-AC Challenge, per-platform UI drive) is a registry entry; a transient crash (rate-limit, socket-close, a device drop §11.4.144) flips it to `crashed`, keeps it OPEN, and respawns until `complete`. The session's done-condition is **not** satisfied while any entry is non-`complete` — a registry showing `complete` without landed evidence is a §11.4 PASS-bluff at the agent-lifecycle layer.
- **Self-meta-test (§11.4.32):** HelixQA plants a known-broken AC and asserts the session reports FAIL — proving the orchestrator itself does not bluff.

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3. Fixtures — real system, autonomous drive, captured wire evidence

Fixture	Real or mock	Notes
the full stack (control plane + edge + client + apps)	**real**	HelixQA drives the deployed system, not a mock
infra (PG/Redis)	**real**, via `containers`	rootless on-demand boot
netns rig	**real**	the E2E substrate (e2e.md)
UI drive	**real** via `panoptic`	the app's own UI; §11.4.117 pixel oracle for non-introspectable surfaces

Fixture	Real or mock	Notes
frame analysis	**real** `vision_engine` OCR	golden-good/bad self-validated (§11.4.107(10))
`.env` credentials	real token, **out-of-band**	§11.4.10 — the one permitted human step
mocks / manual steps	**FORBIDDEN** (§11.4.27 / §11.4.98)	HelixQA is the autonomous-session layer; a human step breaks it

HelixQA's autonomy is the §11.4.98 invariant at the orchestration layer: a session that requires a human to type, click, or trigger anything after startup is a PASS-bluff. A flow genuinely infeasible to drive autonomously is an honest `operator_attended` SKIP with a tracked migration item (§11.4.52), surfaced on the sync channel — never a faked PASS.

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4. Evidence taxonomy — the §11.4.116 sync channel + per-check wire evidence

HelixQA's distinguishing evidence is the **real-time sync channel** plus a **captured wire artifact per check**. Per §11.4.116 the channel is two parts:

1. an **append-only JSONL event stream** (`qa-results/helix_qa/events.jsonl`), one event per line, never rewritten;
2. an **atomically-rewritten status snapshot** (`qa-results/helix_qa/status.json`, write-temp-then-rename) the conductor tails.

```
// qa-results/helix_qa/events.jsonl — append-only (§11.4.116)
{"ts":"2026-…","ev":"session_start","suite":"mvp-dod","build":"<artifact-md5>"}
{"ts":"…","ev":"challenge_start","id":"HVPN-CHAL-AC2-authorized-reach"}
{"ts":"…","ev":"evidence","id":"HVPN-CHAL-AC2-…","path":"qa-results/.../reach.pcap"}
{"ts":"…","ev":"verdict","id":"HVPN-CHAL-AC2-…","result":"PASS","evidence":"qa-results/.../reach.pcap"}
{"ts":"…","ev":"agent","id":"ui-drive-ios","state":"crashed","reason":"device_dropped"}   // §11.4.147
{"ts":"…","ev":"agent","id":"ui-drive-ios","state":"respawned"}
```

The verdict-carries-evidence-path invariant (§11.4.116): a `verdict` event **must** carry the `evidence` path that backs it — a PASS event with no evidence path is a channel-layer PASS-bluff; a snapshot reporting PASS while the stream shows no `evidence` event for that AC is a **contradiction** → treated as FAIL; an AC with no `challenge_start` event cannot have a `verdict`. Verdicts use the closed vocabulary PASS / FAIL / SKIP / OPERATOR-BLOCKED (§11.4.45). The captured artifacts themselves are the §11.4.69 sink-side evidence per check (challenges.md §4).

The status snapshot:

```
// qa-results/helix_qa/status.json — atomically rewritten (write-temp-then-rename)
{ "session":"mvp-dod","build":"<artifact-md5>","phase":"running",
  "counters":{"pass":5,"fail":0,"skip":0,"operator_blocked":0,"pending":4},   // total == 9 ACs (AC1–AC9)
  "last_verdict":{"id":"HVPN-CHAL-AC6-revoke","result":"PASS","evidence":"qa-results/.../revoke_timing.csv"},
  "open_agents":[{"id":"ui-drive-console","state":"in-flight"}] }   // §11.4.147 — not done while open
```

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5. Determinism — N-iteration identical evidence

Per §11.4.50 a HelixQA session over the same artifact MD5 + same rig must produce an identical *verdict set* with structurally-identical per-check evidence across N=3 (normal) / N=10 (cycle-validation):

1. The orchestrator runs under the FA wrapper (full-automation.md §5); each AC's matched-evidence structural hash is compared across runs; a divergence is auto-FAIL.
2. The sync-channel **event ordering** is deterministic per the risk-ordered schedule (§11.4.132 — security-floor ACs first); a reordered run with the same verdicts is acceptable, a *divergent verdict* is not.
3. Crash-respawn does not break determinism: a respawned sub-run must reach the **same** verdict + same evidence shape as a first-attempt run; the registry proves the work was not lost, the evidence proves it was not corrupted (§11.4.147 honest boundary — durability, not correctness; correctness is the per-check evidence).

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6. Acceptance gate — when HQA blocks a release

Gate	Bar	Layer
`make qa` (overview §9)	the `mvp-dod` session reports all 9 ACs PASS with per-check evidence	release sweep

Gate	Bar	Layer
`CM-AUTONOMOUS-FRAMEWORK-SYNC-CHANNEL` (§11.4.116)	the session emits the JSONL stream + atomic snapshot; every verdict carries an evidence path	pre-build + runtime
`CM-HQA-NO-COMPLETE-WITHOUT-EVIDENCE` (§11.4.147)	the done-condition is unsatisfied while any registry entry is non-`complete`; no `complete` without landed evidence	runtime
`CM-HQA-SELF-META-TEST` (§11.4.32)	the planted known-broken AC makes the session report FAIL	pre-build meta-test

HelixQA's session exit status **gates the release** — `make qa` is the §11.4.40 pre-tag full-suite-retest driver at N=10, and the §11.4.165 independent-verification gate (QA-D4: **independent re-read**, overview §10 — `vision_engine` re-reads the MP4 and cross-checks the core FSM, the harness's claim is never the verdict). A finding generates a workable-items ticket (§11.4.148); the feature's coverage-ledger cell stays out of `AUTONOMOUS_VERIFIED` until the session re-runs GREEN.

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7. The paired §1.1 mutation (anti-bluff proof)

```
# §1.1 mutation for CM-AUTONOMOUS-FRAMEWORK-SYNC-CHANNEL (verdict-carries-evidence)
- make the orchestrator emit a verdict event WITHOUT an `evidence` path:
    {"ev":"verdict","id":"HVPN-CHAL-AC2-…","result":"PASS"}     // no evidence path
- assert: the channel validator flags a PASS-with-no-evidence-path as a contradiction
  → treats it as FAIL → gate FAILs (the bluff is caught)
- restore (evidence path required); assert PASS

# §1.1 mutation for CM-HQA-NO-COMPLETE-WITHOUT-EVIDENCE (§11.4.147)
- mark a `crashed` registry entry as the session's done-condition `complete` WITHOUT
  its sub-run landing evidence
- assert: the done-condition checker refuses to read satisfied → session reports
  incomplete → gate FAILs (crash≠done enforced)
- restore (respawn-until-real-complete); assert the session completes only with evidence

# §1.1 mutation for CM-HQA-SELF-META-TEST (§11.4.32 — the orchestrator itself doesn't bluff)
- plant a known-broken AC (e.g. force default-deny to fail-open in the test build)
```

```
- assert: the mvp-dod session reports AC3 FAIL (the orchestrator
detects the break)
- a session that reports AC3 PASS on the broken build is itself
the bluff → meta-test FAILs
```

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8. Test skeletons

8.1 The orchestrator invocation + sync-channel consumption

```
// submodules/helix_qa/cmd/orchestrator/main.go (invocation
        contract; engine in submodule)
//   go run ./submodules/helix_qa/cmd/orchestrator --suite mvp-dod
// 1. boot infra via containers (§11.4.76), bring up the rig
        (§11.4.3)
// 2. for each AC in the mvp-dod bank, dispatch a sub-run
        (registry entry §11.4.147)
// 3. score via the challenges engine (§5.5); drive UI via
        panoptic; analyse via vision_engine
// 4. append every event to events.jsonl; rewrite status.json
        atomically (§11.4.116)
// 5. on crash → flip registry to `crashed`, respawn until
        `complete`
// 6. session PASS only when every AC verdict==PASS AND every
        registry entry==complete-with-evidence

# the conductor's live tail of the sync channel (§11.4.116) —
        never idle-blind (§11.4.94)
tail -F qa-results/helix_qa/events.jsonl | while read -r ev; do
  case "$(jq -r .ev <<<"$ev")" in
    verdict)
      path=$(jq -r '.evidence // empty' <<<"$ev")
      [ -n "$path" ] || ab_fail "§11.4.116: verdict with no
        evidence path → contradiction (FAIL)"
      ;;
    agent) [ "$(jq -r .state <<<"$ev")" = crashed ] &&
        note_open_workunit "$ev" ;;  # §11.4.147
  esac
done
```

8.2 A control-plane bank suite entry (drives INT + CHAL)

```
# helix_qa/banks/control_plane.yaml — written test bank per
        service (§11.4.27)
bank: helixvpn-control-plane
suites:
  - name: rls-isolation
    drives: [ "go test -tags integration -run
        TestRLSCrossTenantDenial ./helix-go/internal/store/..." ]
```

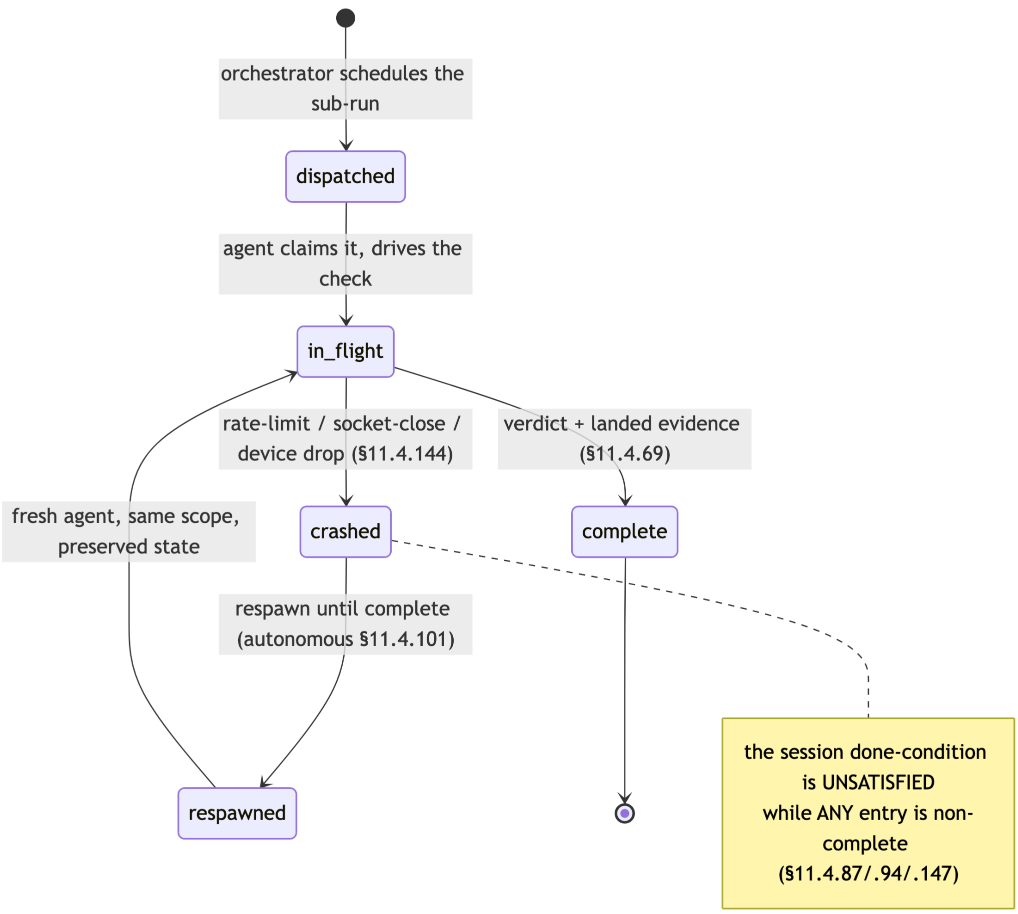
```yaml
    evidence_class: rls_rowset                  # §11.4.69 —
        captured rowset
    capture: qa-results/int/rls_${RUN_ID}.json
  - name: revoke-latency
    drives: [ "rig/revoke.sh" ]
    evidence_class: counter_delta
    assert_p99_lt_seconds: 1.0                   # SLO3
    capture: qa-results/e2e/revoke_timing.csv
```

8.3 A per-platform client bank suite (drives UI + REC, §11.4.165 independent re-read)

```yaml
# helix_qa/banks/clients_ios.yaml — written bank per platform
        (§11.4.27)
bank: helixvpn-client-ios
suites:
  - name: connect-flow
    drives: [ "panoptic drive --app access --platform ios --flow
        enroll-connect-reach" ]
    evidence_class: REC                          # §11.4.153/.159
    capture:
      kind: mp4
      window_scoped: true                        # §11.4.154 — app
        window only
      filename_prefix: helixvpn---               # §11.4.155
      vision_verdict: vision_engine              # §11.4.163
        independent re-read (§11.4.165)
      expected_patterns:
        ["Connect","Connecting","Connected","shield:green"]
      cross_check: { core_fsm_at_green_shield: "Connected" }   #
        defeats B3
    on_uninstrumentable_ui: pixel_oracle         # §11.4.117
        fallback, never fake PASS
    on_no_device: skip_operator_attended         # §11.4.3/.52
        honest SKIP (real device required)
```

8a. The agent-lifecycle registry (§11.4.147 — crash ≠ done)

Every HelixQA sub-run (a per-AC Challenge, a per-platform UI drive, a device session) is a durable registry entry on the §11.4.116 sync substrate, so a crash never loses, forgets, or corrupts its work:



Lifecycle hazard	HelixQA response
transient API rate-limit kills a sub-run	flip to `crashed`, keep OPEN, respawn with the project's defined backoff (never invented, §11.4.6)
a tracked device drops mid-UI-drive	§11.4.144 availability-following: log an honest offline event, wait the defined reconnect budget, re-attach + resume, escalate on timeout
partial UI-drive state (half a recording)	preserve, §11.4.84 quiescence check, resume-or-clean-restart from a known-good base
a `complete` entry has no landed evidence	the done-condition refuses to read satisfied (a `complete` without evidence is a §11.4 PASS-bluff)

The honest boundary (§11.4.147): the registry + respawn guarantee the work is **not lost or corrupted**, NOT that it is **correct** — a respawned sub-run's output still crosses the per-check §11.4.69 evidence gate and the §11.4.40 retest; §11.4.147 is the durability layer beneath those.

8b. HelixQA as the independent verifier (§11.4.165) — the structural seam

HelixQA is dispatched as a **structurally-separate agent from the code author** (§11.4.70 / §11.4.165); §11.4.92 author-side self-evaluation PRECEDES but never satisfies it. The independence is load-bearing:

Verification axis	What HelixQA independently re-reads	The author claim it does NOT trust
reachability	the captured pcap (SYN-ACK present/absent)	"the reach test passed"
escalation	the tshark classification (H3 present, WG absent)	`StatusReport.transport=="masque-h3"` alone (B3)
UI flow	the window-scoped MP4 via `vision_engine` + core-FSM cross-check	"the shield went green" (a green shield while the FSM is `Blocked` FAILs)
no-log	the live-DB introspection report	"the schema is fine"

A HelixQA session that rubber-stamps the harness's claims without re-reading the artifacts is itself the bluff the §11.4.165 + §11.4.32 self-meta-test (§7) catches.

9. Open decisions surfaced for QA

#	Decision	Options	Recommendation
H-D1	Vision verdict authority (QA-D4)	trust the harness verdict vs independent `vision_engine` re-read	**independent re-read** (overview §10) — `vision_engine` re-reads the MP4 + cross-checks the core FSM (defeats B3); the harness claim is never the verdict (§11.4.165)
H-D2		fixed vs project-defined budget	**project-defined backoff (§11.4.147)** reusing the

#	Decision	Options	Recommendation
	crash-respawn backoff		recovery path's existing budgets — never invented numbers (§11.4.6)
H-D3	per-platform bank granularity	one bank per platform vs one bank with platform matrix	**one bank per platform** for clear per-surface coverage (§11.4.27); a matrix view is a derived projection

> **`UNVERIFIED:`** the `helix_qa` orchestrator CLI (`--suite`), the bank-YAML keys (`drives`, `evidence_class`, `vision_verdict`, `on_no_device`), and the `cmd/orchestrator` path are the **intended** shape per the overview skeleton; confirm against the pinned `HelixDevelopment/helix_qa` SHA (§11.4.74 — extend upstream if unsupported).

Sources verified

- Volume-8 overview §2 (taxonomy row `HQA`), §5.6 (the orchestrator + the §11.4.116 sync-channel JSONL skeleton), §5.5 (banks it consumes), §7.2 (AC set it scores), §10 QA-D4 — read 2026-06-26.
- Sibling specs cross-referenced: challenges.md, e2e.md, integration.md, full-automation.md — written 2026-06-26 in this same volume.
- Constitution: §11.4.27 (no-fakes / 100% type coverage / HelixQA banks + autonomous sessions), §11.4.116 (real-time sync channel / verdict-carries-evidence-path), §11.4.147 (crashed-agent respawn / crash≠done registry), §11.4.165 (independent verification agent), §11.4.32 (self-meta-test), §11.4.45 (closed verdict vocabulary), §11.4.69 (sink-side evidence per check), §11.4.98 (autonomous / no manual step), §11.4.117 (pixel-oracle fallback), §11.4.52/.3 (honest operator-attended SKIP), §11.4.153/.154/.155/.159/.163 (REC / window-scope / prefix / media-validation), §11.4.148 (ticket integrity), §11.4.93/.95 (workable-items DB), §11.4.50 (determinism), §11.4.40 (pre-tag retest), §11.4.144 (device-availability following), §1.1 (paired mutation) — from `CLAUDE.md` in-context.
- The `helix_qa` orchestrator CLI + bank-YAML schema are reproduced from the overview skeleton — **not independently fetched from `HelixDevelopment/helix_qa`** (`UNVERIFIED`; confirm at the pinned SHA).